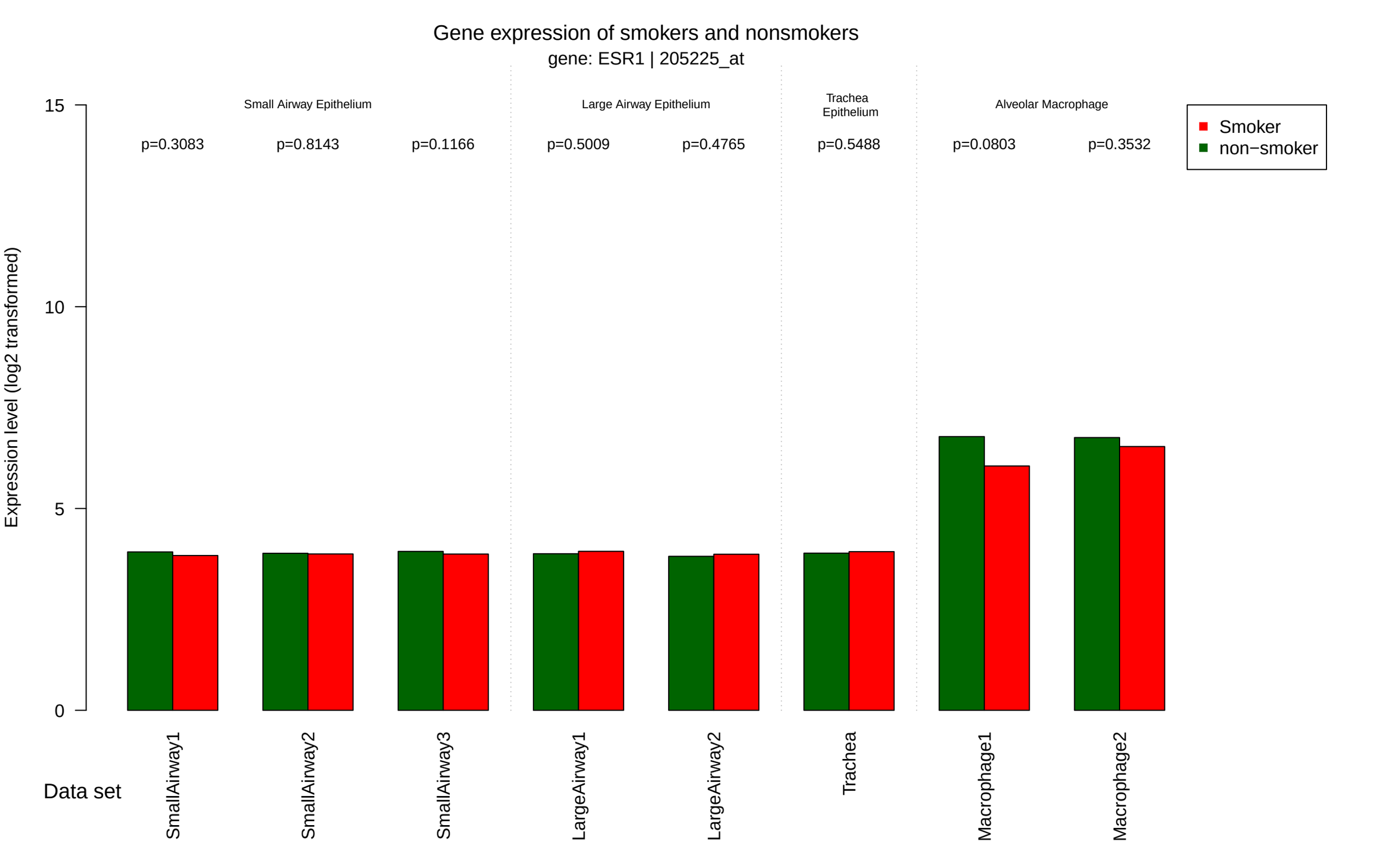
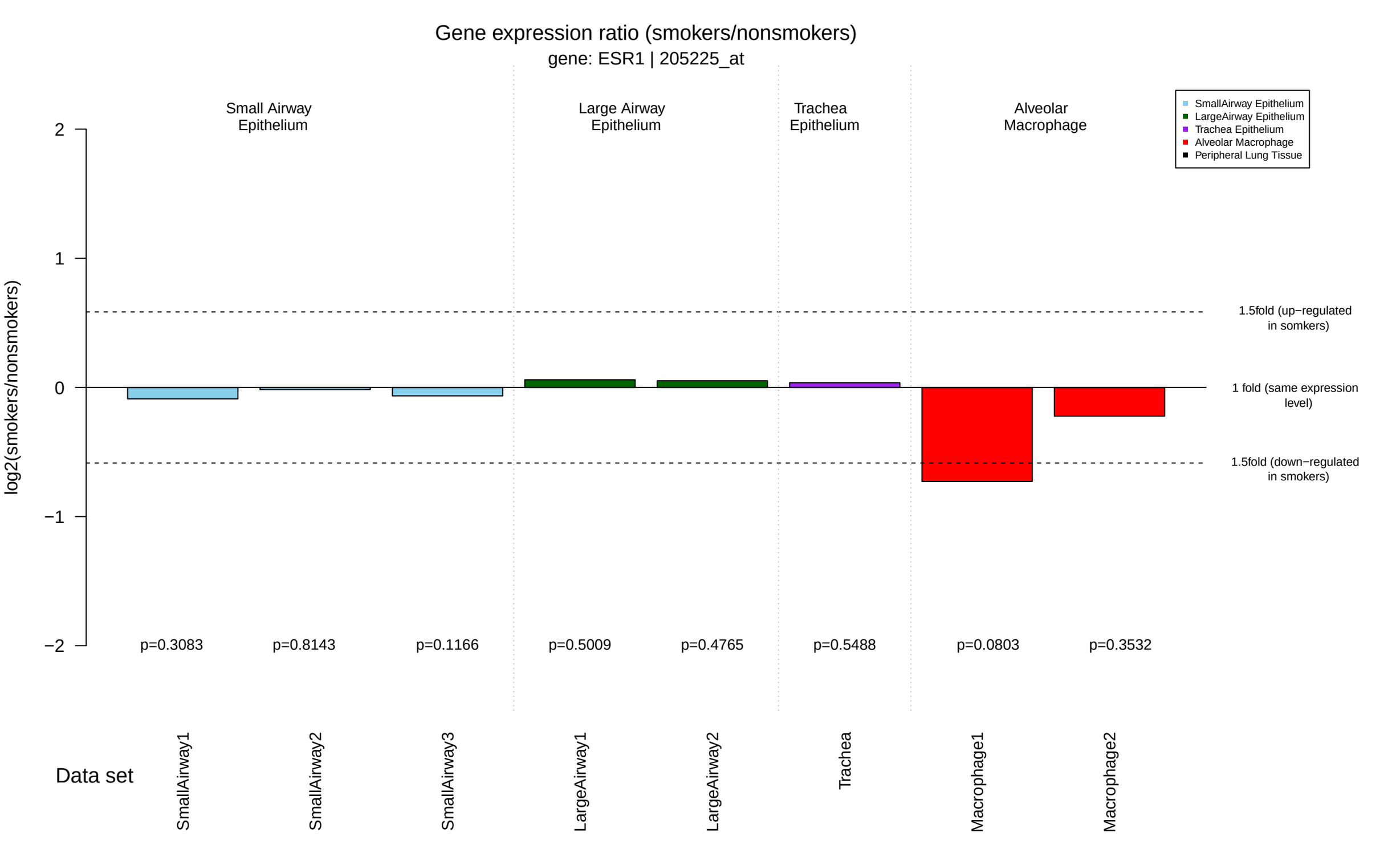
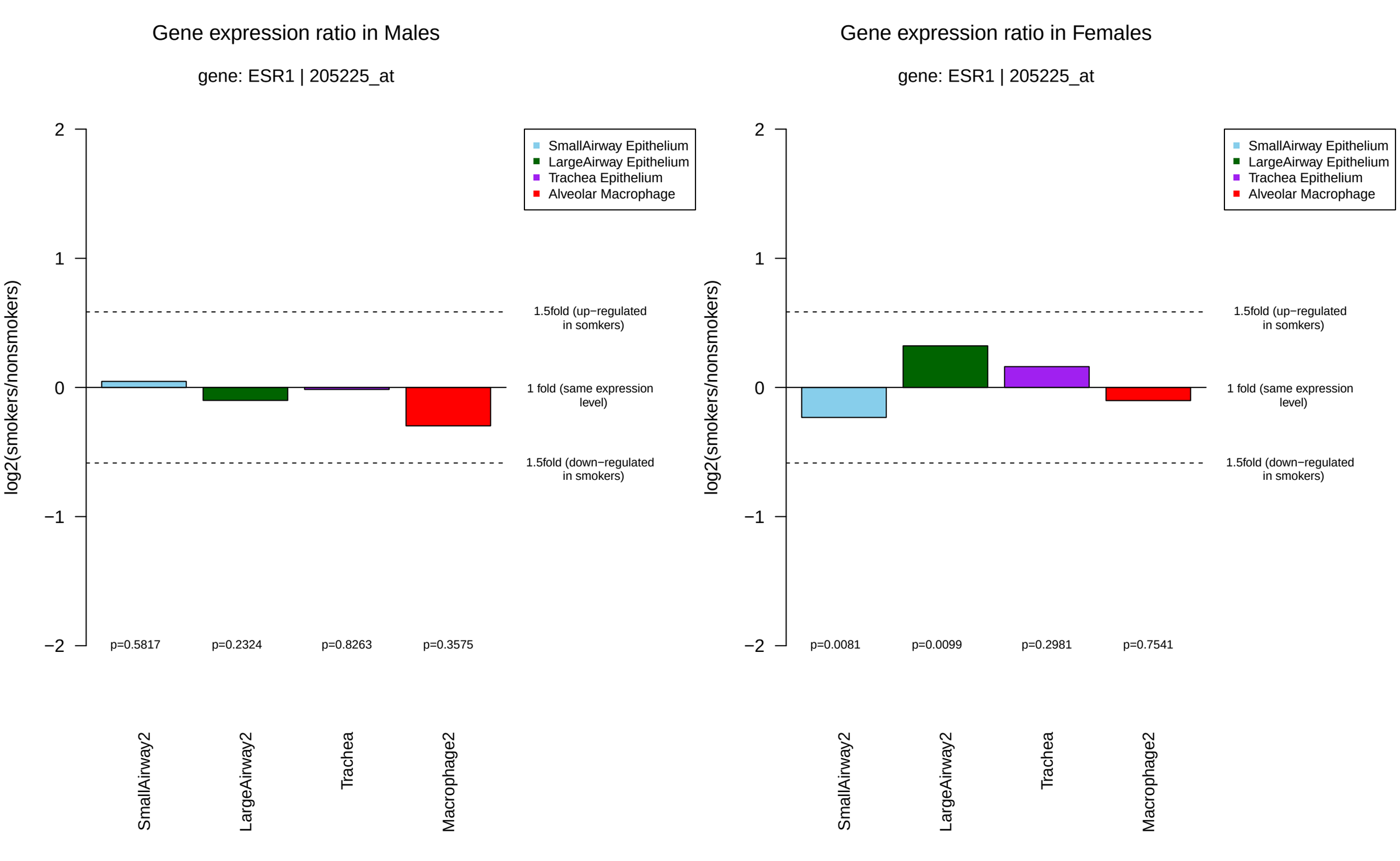
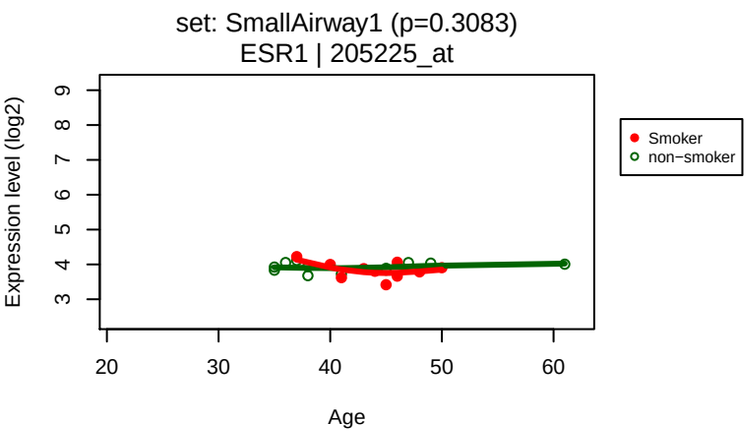
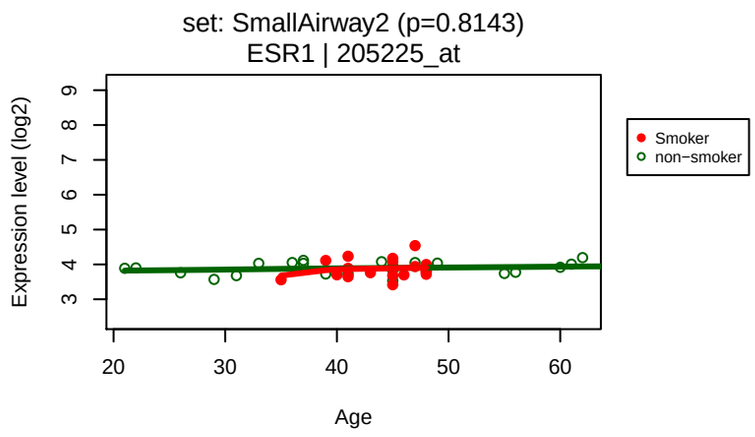
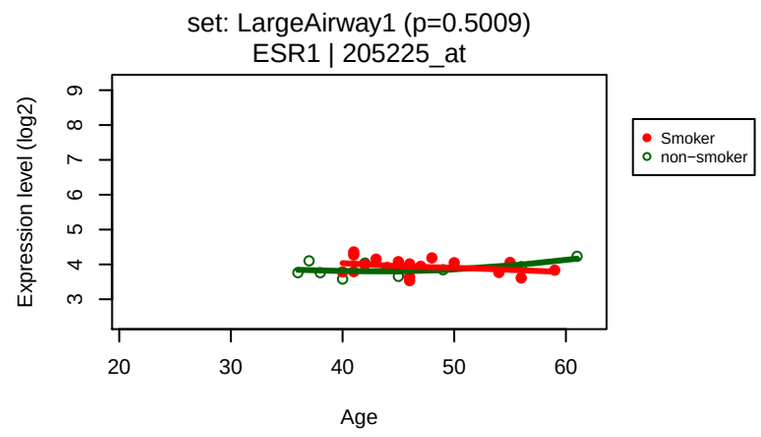

Gene expression of smokers and nonsmokers
gene: ESR1 | 205225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3083
p=0.8143
p=0.1166
p=0.5009
p=0.4765
p=0.5488
p=0.0803
p=0.3532
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

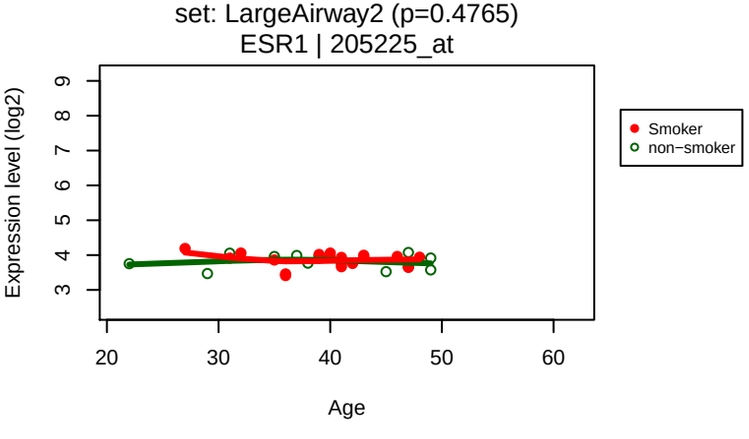

Gene expression ratio (smokers/nonsmokers)
gene: ESR1 | 205225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3083
p=0.8143
p=0.1166
p=0.5009
p=0.4765
p=0.5488
p=0.0803
p=0.3532
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

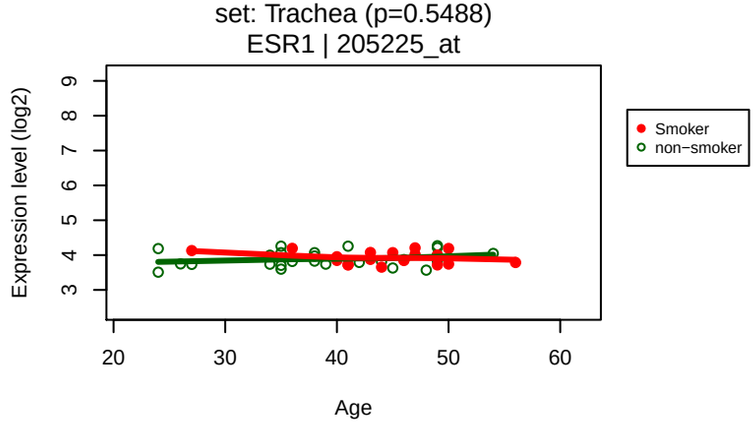

Gene expression ratio in Males
gene: ESR1 | 205225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5817
p=0.2324
p=0.8263
p=0.3575
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ESR1 | 205225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0081
p=0.0099
p=0.2981
p=0.7541
SmallAirway2
LargeAirway2
Trachea
Macrophage2

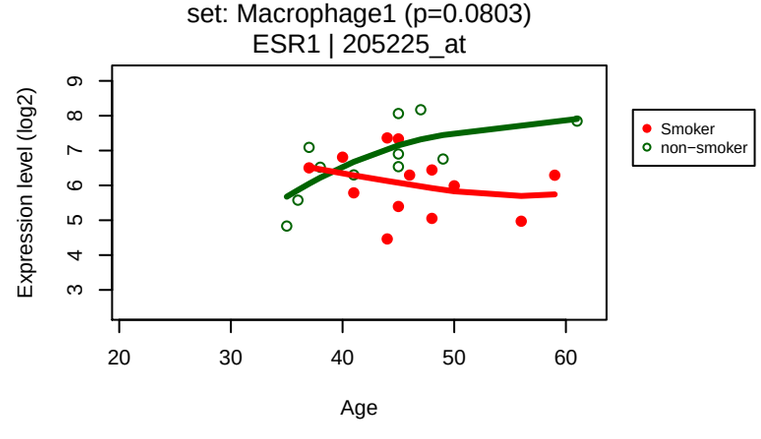

set: SmallAirway1 (p=0.3083)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker

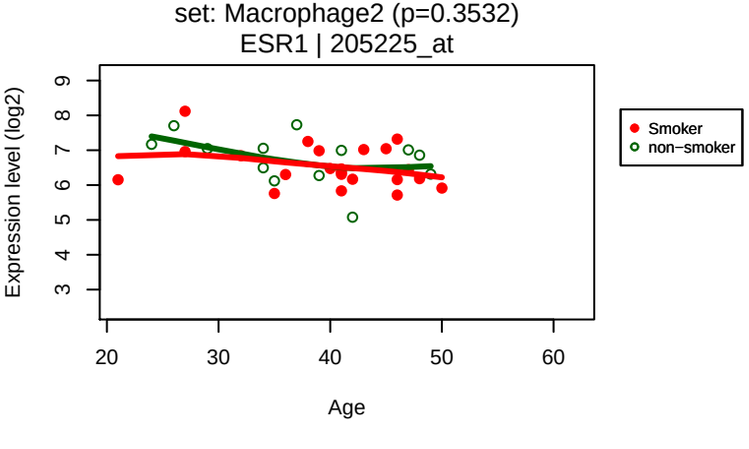

set: SmallAirway2 (p=0.8143)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker

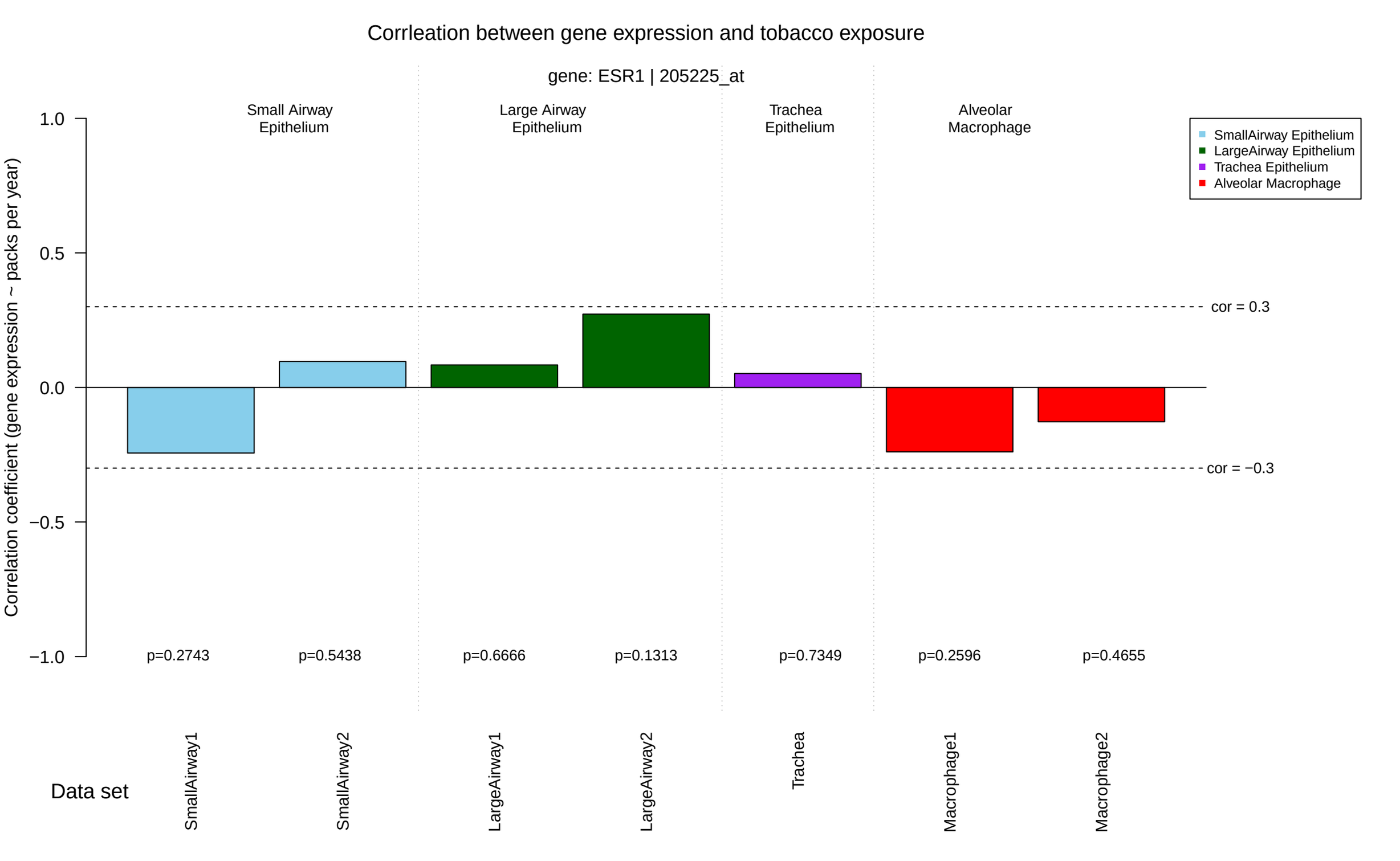

set: LargeAirway1 (p=0.5009)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4765)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5488)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0803)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3532)
ESR1 | 205225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ESR1 | 205225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2743
p=0.5438
p=0.6666
p=0.1313
p=0.7349
p=0.2596
p=0.4655
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2